Grid WAR ToDo List

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- From Dan Kahan: I noticed some glitches in the data, and to try to keep your ”to do” list from expanding or at least to make it easier to finish off, I tried to identify & fix some of them.
  - Simplest was a conflicting mix of team identifiers drawn from from retrosheet and different source (not FG or at least not only FG; maybe Baseball Savant?). E.g., ”NMY, NYN,” ”NYA, NYY,” ”Ariz, AZ,” ”SF, SFN,” ”LAN, LAD,” ”SDN, SD,” ”CNH, CHC,” ”CWS, CHA,” etc. I was easily able to identify and fix those.
  - I also thought I’d just match the Grid War 1952-2025 data w/ retrosheet to add the pitcher identifiers for you. That’s where I ran into some more difficult problems. Basically, there are a bunch of discrepancies. They include: 20 duplicate rows, 473 misidentified starters, 906 missed regular season starts
  - When I first tried to figure this out, I noticed that some of the misidentified starters were pitchers who entered a game after the starter was removed w/ 0 batters faced. MLB rules still treat the pitcher who left under those circumstances as the ”starter” but perhaps you might want to classify him otherwise. There are about 50 of those–w/ a column to id them.
  - Also I think some fraction of the misidentified starters likely arose from a coding glitch about how to identify the second starting pitchers in double headers; this was something that I just noticed casually, and didn’t try to fix. I don’t think this issue covers all the remaining misidentifications, though.
- Fix: use pitcher ID, not pitcher name, to index pitchers. It only lists one pitcher per name per season in such cases and might in fact be conflating them. E.g. Bobby Jones of NYN is identified with 50 (!) starts in 1998, whereas in fact he had 30 & COL’s Bobby Jones, who is omitted from file, started 20 games; in 1999, COL Bobby Jones is listed as starting 29, when in fact he started 20 and NYM’s Jones, now unlisted, started 9.
- **Defining Grid WAR when a pitcher goes longer than 9 innings**
  - ◇ Currently we evaluate such pitchers by taking their runs allowed and truncating their IP at 9. This is not correct.
  - *From Adam Brodie*: For example, we should give Cliff Lee credit for his scoreless 10^{th} inning. The 96.5% chance of the team winning after holding the opponent scoreless through 9 breaks out into something like 95% chance of winning in 9 (made up arbitrary number), plus a 3% chance the game is in a scoreless tie after 9 multiplied by a 50% chance that the team wins in extra innings (I understand the numbers don’t add up here, I appreciate this is a quirk of the Poisson approximation, we can ignore that, for purposes of

discourse it's more useful to preserve an intuitive 50% WPCT in extra innings than to preserve consistent outcome rates for games where the opponent didn't score through 9) . Now, Lee getting through the 10th scoreless is informative; his team would now be more likely than 50% to win in extra innings, say it's 30% in 10 and then 50% of the complementary 70% Lee's offense doesn't come through immediately, coming out to 65%. That would suggest that rather than the .95 + .5 x .03 = .965 value, his performance would be worth .95 + .65 x .03 = .970. And we can see how this would continue to grow towards some limit should the game have continued to remain scoreless; after 9 innings 97% of instances are determined and the majority of those are wins for Lee's team, 3% of games are undecided, then for each successive scoreless inning Lee records, some portion of the undecided instances get whittled away, with some more wins going to Lee's count.

	Year	Date	Game Type	Pitcher	Team	GWAR	GWAR+	Home Team	Away Team	Off. Factor
4504	2012	2012-04-18	Reg	Cliff Lee	PHI	0.519	0.519	SF	PHI	0
4524	2012	2012-04-18	Reg	Matt Cain	SF	0.519	0.51	SF	PHI	-0.009

- *From Adam Brodie*: Famous game, Harvey Haddix perfect through 12 game, 5/26/1959, Harvey Haddix allows 1 against MLN over 12.2, Lew Burdette goes 13 scoreless against PIT. So this is one where Haddix actually appears to be penalized for his team's inability to score before extra innings, something we're very explicitly trying to avoid in this framework. If we wanted to truncate the performance at 9 innings, which is a very reasonable approximation, Haddix should be credited with having held his opponent scoreless to that point. His context neutral game-performance win probability is 86.3% compared to Burdette's 96.1%. Based on the plots I'm inferring that these refer to the win probabilities associated with allowing 1 and 0 runs, respectively, through 9 innings. Seems like the sensible solution here in the absence of laborious machinery to accommodate extra inning performance is to calculate Haddix's performance through 9 IP explicitly rather than to take his RA from his game line and truncating his IP to 9. Given that you're already reporting game exit conditions I assume this is actually a relatively painless extra amount of calculation.

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- **Improving the f grid**
  - Allow different Poisson parameters for each inning. For instance, the first inning should have a higher λ since better batters bat at the top of the lineup.
  - Allow different Poisson parameters for each inning depending on when a starting pitcher is pulled. In particular, middle relievers tend to be worse than starting pitchers, suggesting a higher value of λ for those innings. Closers are often very good pitchers, suggesting a lower value of λ.
  - The distribution of runs scored in a half-inning is not Poisson; more likely it is a zero-inflated Poisson, a more general Conway-Maxwell-Poisson, or a similar distribution on the non- negative integers.

- **Adjust for opponent quality.** Adjust for opponent quality using the full batting lineup of the opposing team, rather than a fixed constant for each team-season.
- **Adjust for fielding.** Perhaps use a version of expected runs allowed that adjusts for fielding, rather than runs allowed, as a base measure of pitcher performance.
- *From Pat:* I believe you are combining two (or more) pitchers with the same name – who happened to have starts in the same season – under one individual. I noticed it when I saw that Bobby Jones-1998 had 50 starts. (Bobby Jones, the Met had 30 and Bobby Jones the Rockie had 20). We've got two starters named Pedro Martinez from 1994 and I don't know how many other cases of this. Several years ago, this got to be confusing for me (2 Bill Swifts, 2 Dutch Leonards, 2 Bill Lees, 2 Bob Gibsons, etc.)... and that's when I began using the unique Retrosheet played IDs to keep things straight.
- $w_{\mathbf{rep}}$ **should vary by inning.**
  - A starter who goes 1 inning should be compared to a replacement-level starter who goes 1 inning. That is because the long relievers who replace the innings that would've been taken by a starter who went longer will be better than a replacement-level starter. We can deal with this by using a different poisson λ in each inning depending on the quality of pitcher (e.g., one for replacement-level, one for long reliever who enters early in the game, etc.)
  - *From Matan K and Hareeb al-Saq:* Wouldn't this suggest that wrep is 0.43 (43%) for using a replacement-level pitcher for 1 IP? It's using a wrep value for a replacement pitcher throwing an entire start instead of just 1 inning. Grid WAR (though not the "Grid WAA" framework) would be biased for SP that pitch less in a given start, with a potential downstream effect on the conclusion re: mediocre pitchers, though I may definitely be wrong. Agreed. It's comparing to 0.428 right now regardless of IP, and the number should start at 0.5 (ignoring H/A) and go down with each IP. And since mediocre pitchers average fewer IP/start on average than good pitchers (in aggregate), that formula overvalues them (in aggregate).